University of Science and Technology of Hanoi

Fundamentals of Data Science

USA Flight Delay Analysis

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ABSTRACT

Flight delays pose a significant operational and economic challenge for the aviation industry, affecting both airline efficiency and passenger experience. This project examines the factors influencing flight departure delays by integrating historical flight records with meteorological data for major airports in Washington State. A data engineering pipeline was developed to collect, clean, and merge data from the Bureau of Transportation Statistics (BTS) and the National Oceanic and Atmospheric Administration (NOAA). After comprehensive preprocessing and feature engineering, several machine learning models were employed to analyze delay patterns and identify key predictors. Statistical analysis and visualization techniques were applied to explore relationships between time, weather, and route characteristics under different delay scenarios. The findings provide insights into the underlying dynamics of flight delays and demonstrate how data-driven approaches can enhance predictive capability and operational decision-making in the aviation sector.

Keywords: *Flight Delay Prediction, Data Engineering, Machine Learning, Weather Integration, Statistical Analysis, Feature Engineering, Aviation Analytics.*

Chapter 1

Introduction

1.1 Business Analysis

1.1.1 Background

Flight delays are a persistent challenge in the aviation industry, leading to significant operational, financial, and customer-service impacts. Even minor delays can cascade across the network, disrupting schedules, increasing costs, and reducing passenger satisfaction. With the expansion of air traffic and the growing influence of external factors such as weather and airport congestion, understanding and predicting flight delays has become an essential task for both airlines and transportation authorities. The integration of data science and machine-learning techniques provides new opportunities to analyze historical data and uncover complex relationships between flight, weather, and operational variables.

1.1.2 Problem Statement

Flight operations are influenced by multiple interacting factors—weather conditions, scheduling patterns, route distances, and airline operations. Traditional statistical approaches often fail to capture the non-linear dependencies and dynamic patterns that drive delay events. This study aims to investigate the key factors contributing to flight departure delays and to develop a data-driven analytical framework that can detect patterns and relationships across diverse datasets. By merging flight performance data with meteorological information, the study seeks to identify major predictors of delay and examine how environmental and temporal factors affect flight punctuality.

1.1.3 Analytical Approach

This project employs a data engineering and machine-learning pipeline to process and analyze flight and weather data. Datasets from the Bureau of Transportation Statistics (BTS) and the National Oceanic and Atmospheric Administration (NOAA) were collected, cleaned, and merged through automated data-processing steps. Feature engineering was applied to transform time-based and weather-related variables into model-ready formats. Descriptive statistics, correlation analysis, and visualization techniques were used to explore relationships among variables, while classification algorithms such as Logistic Regression, Random Forest, and LightGBM were implemented to model delay likelihood. Insights derived from these analyses contribute to understanding the determinants of flight delay behavior.

1.1.4 Goals

The goal of this study is to build a comprehensive analytical system capable of predicting flight departure delays and identifying their main causes. Specifically, it aims to integrate flight and weather data into a unified dataset, construct machine-learning models to analyze delay patterns, and generate interpretable insights that can support airline operations and scheduling decisions. Through this approach, the project bridges data engineering and data science to enhance the reliability and efficiency of air-transport services.

1.1.5 Implications

The outcomes of this analysis can assist airlines, airports, and regulators in improving operational planning and resource allocation. By recognizing weather-related and temporal factors that most strongly affect delay risk, stakeholders can design early-warning systems, optimize flight scheduling, and enhance passenger communication. Moreover, the integration of predictive modeling into decision-support systems demonstrates how data-driven methods can contribute to more resilient and efficient aviation management.

Chapter 2

Data Collection and Preparation

2.1 Introduction

This project employs two distinct datasets — one containing flight operation records and another containing hourly weather observations — to support the analysis and prediction of flight departure delays. The use of both datasets is essential because flight performance is not only influenced by operational factors such as scheduling and routes, but also by environmental conditions like temperature, wind, and visibility. By combining these two perspectives, the study aims to capture a more realistic representation of the factors contributing to flight delays.

Therefore, each dataset was processed and cleaned individually before being merged into a single analytical dataset that serves as the foundation for further exploration and modeling.

2.1.1 Flight Dataset

The flight dataset used in this project is the **Airline On-Time Performance Data**, obtained from the **U.S. Bureau of Transportation Statistics**, which provides detailed records of domestic flight operations across the United States. It contains information about each scheduled flight, including the airline carrier, origin and destination airports, scheduled and actual departure times,... This dataset serves as the primary source for analyzing operational factors contributing to departure delays.

For this study, only flights departing from Washington State during the second quarter (Q2) were used. The dataset consists of 65,374 entries and 17 features, and is provided in CSV (Comma-Separated Values) format.

MONTH	DAY_OF_MONTH	OP_UNIQUE_CARRIER	ORIGIN	CRS_ARR_TIME	DEST	...	DISTANCE	DEP_DELAY	DEP_DEL15
4	1	AA	SEA	9.33	SEA	...	2279.0	11.0	0
4	1	UA	GEG	18.12	LAS	...	224.0	18.0	1
4	1	AS	SEA	15.32	BUR	...	937.0	20.0	1
...	...	...	...	...	...	...	...	...	...
6	30	WN	SEA	22.75	SMF	...	605.0	-5.0	0
6	30	WN	SEA	16.92	STL	...	1709.0	10.0	0

Table 2.1: Airline On-Time Performance Dataset

The data types and detailed discussion of each feature in the dataset are provided below:

Features	**Type**	**Data Type**	**Explanation**
MONTH	Temporal	Integer	Month of the flight date.
DAY_OF_MONTH	Temporal	Integer	Day of the month.
DAY_OF_WEEK	Temporal	Integer	Day of the week.
OP_UNIQUE_CARRIER	Categorical	String	Unique carrier code identifying the airline.
ORIGIN	Categorical	String	Code of the departure (origin) airport.
ORIGIN_STATE_ABR	Categorical	String	Two-letter abbreviation of the origin state.
DEST	Categorical	String	Code of the arrival (destination) airport.
DEST_STATE_ABR	Categorical	String	Two-letter abbreviation of the destination state.
CRS_DEP_TIME	Temporal	Integer	Scheduled departure time (in HHMM format).
CRS_ARR_TIME	Temporal	Integer	Scheduled arrival time (in HHMM format).
CANCELLED	Quantitative (Binary)	Float	Indicates if the flight was cancelled (1 = Yes, 0 = No).
CRS_ELAPSED_TIME	Quantitative	Float	Scheduled flight time in minutes.
DISTANCE	Quantitative	Float	Distance between origin and destination airports (in miles).
DISTANCE_GROUP	Categorical (Ordinal)	Integer	Group number representing distance range.
DEP_DELAY	Quantitative	Float	Difference between actual and scheduled departure time.
DEP_DEL15	Quantitative (Binary)	Integer	1 if departure delay ¿ 15 minutes, otherwise 0.
DEP_DELAY_GROUP	Categorical (Ordinal)	Float	Grouping of departure delay intervals.

Table 2.2: Features Explanation

2.1.2 Weather Dataset

The weather dataset used in this project was obtained from the **National Oceanic and Atmospheric Administration (NOAA)**, which provides historical hourly weather observations recorded at airports across the United States. To ensure consistency with the flight dataset, only data from major airports in Washington State during the second quarter (Q2) were collected. The dataset contains over 50,000 hourly records with many key meteorological, stored in CSV format, with timestamps adjusted to local time zones to enable accurate integration with flight schedules.

STATION	**DATE**	**HourlyDewPointTemperature**	**HourlyDryBulbTemperature**	**HourlyRelativeHumidity**	**HourlyVisibility**	**HourlyWindSpeed**
72784524163	2025-04-01T00:53:00	35	41	79	10	10
72784524163	2025-04-01T01:53:00	34	41	76	10	0
...	...	...	...	...	...	...
72793724222	2025-06-30T23:53:00	51	64	63	10	3
72793724222	2025-06-30T18:53:00	52	69	55	10	9

Table 2.3: NOAA Dataset

The data types and detailed discussion of each feature in the dataset are provided below:

STATION	Categorical	String	Identifier code where the weather data was recorded.
DATE	Temporal	Datetime	Timestamp representing the date and hour of the observation.
HourlyDewPointTemperature	Quantitative	Float	Dew point temperature (°F).
HourlyDryBulbTemperature	Quantitative	Float	Air temperature (°F).
HourlyRelativeHumidity	Quantitative	Float	Percentage of relative humidity in the air (%).
HourlyVisibility	Quantitative	Float	Horizontal visibility distance (miles).
HourlyWindSpeed	Quantitative	Float	Average wind speed during the hour (miles per hour).

Table 2.4: Features Explanation

2.2 Data Preparation and Cleaning

2.2.1 Flight Data Cleaning and Preprocessing

To ensure the data is suitable for analysis and modeling, several preprocessing steps were performed:
Filtering by origin state:

- Kept only flights with ORIGIN_STATE_ABR = 'WA', since the state has five main airports, which simplifies the merging process with weather data.
- After this step, a total of 50372 flights departing from WA during Q2 were recorded.

Handling Missing and Duplicate Records:

- Removed all rows with missing or zero values, except for time-related variables (DEP_DELAY, DEP_DEL15, DEP_DELAY_GROUP, DEP_DELAY_NEW).
- Duplicate records were identified and removed to maintain data integrity.

Remove cancelled flight:

- Filtered out all cancelled flights (CANCELLED = 1) to focus on actual departure delays.
- Remove CANCELLED column.

Feature Engineering:

- CRS_DEP_TIME and CRS_ARR_TIME were converted from HHMM to decimal-hour format.
- New feature ORIGIN-DEST was created to represent flight routes for better analytical insight.

General:

- After prepareing, dataset have 17 features and 50186 samples.
- Dataset is ready for merging and analysis.

2.2.2 Weather Data Cleaning and Preprocessing

The weather dataset was cleaned and standardized to ensure accuracy and consistency before merging with the flight data **Handling Missing and Duplicate Records:**

- Removed records containing NULL values.

- Duplicate records were identified and removed.

Extract new features:

- Converted the DATE column to datetime format to enable time-based operations and ensure consistent timestamp handling.
- Extracted the day and month information into a new column date, formatted as "MM-DD" for easier temporal alignment.
- Extracted the hour component into a new column hour, which was later used to merge weather data with flight departure times.

Map airport codes:

- The STATION codes were mapped from NOAA station IDs to corresponding airport IATA codes to ensure consistency with the flight dataset during merging.

General:

- After prepareing, dataset have 8 features and 10851 samples.
- Dataset is ready for merging and analysis.

2.2.3 Merging Datasets

After cleaning and preprocessing both datasets, they were merged to create a comprehensive dataset for analysis:

Prepare Time Columns:

- Extracted the departure hour (DEP_HOUR) from CRS_DEP_TIME by flooring the scheduled time to the nearest hour.
- Created a new column FL_DATE by combining MONTH and DAY_OF_MONTH, with the year temporarily set to 2025.
- In the weather dataset, added the year "2025" to the date column (if missing) and converted it to datetime format.
- These transformations standardized date and time formats across both datasets, enabling accurate merging by hour and location.

Merge Flight & Weather Data

- Merged flight and weather datasets using a left join on matching keys: ORIGIN, FL_DATE, DEP_HOUR -¿ STATION, date, hour.
- Ensured each flight matched its corresponding hourly weather record.

- Removed intermediate columns that were no longer needed: [”DEP_HOUR”, ”FL_DATE”, ”STATION”, ”date”, ”hour”].
- The merged dataset contained 50,029 rows and 22 columns.

MONTH	DAY_OF_MONTH	ORIGIN	HourlyVisibility	HourlyWindSpeed	...	DEST	DISTANCE	DEP_DELAY	DEP_DEL15
4	1	GEG	5.0	10.0	...	PHX	1020.0	-5.0	0
4	1	GEG	10.0	10.0	...	PHX	1660.0	15.0	1
...	...	...	...	...	...	...	...	...	...
6	30	SEA	10.0	14.0	...	SMF	605.0	-5.0	0
6	30	SEA	10.0	9.0	...	STL	1709.0	10.0	0

Table 2.5: Final Dataset

2.3 Feature Engineering

2.4 Final Dataset Overview

Chapter 3

Exploratory Data Analysis

3.1 Dataset Overview

The cleaned dataset consists of 50,372 flight records collected from airports in Washington State (WA) during the second quarter (Q2) of 2023. After preprocessing and merging with hourly weather observations, the dataset contains 17 variables describing flight operations, scheduling information, and weather conditions.

Data quality checks show that:

- There are no duplicated rows after merging.
- There are no critical missing values in the key predictors or the target variable.

The dataset includes 5 categorical features: OP_UNIQUE_CARRIER, ORIGIN, ORIGIN_STATE_ABR, DEST, and DEST_STATE_ABR. Among them, there are 12 airlines operating flights from 5 origin airports (all within Washington State) to 86 destination airports across 36 states.

3.2 Target and Feature Distributions

3.2.1 Target Distributions

The target variable DEP_DEL15 indicates whether a flight experienced a departure delay.

Out of a total of 45,251 flights, about 9,909 were delayed, while 35,342 departed on time or early.

This shows a moderate class imbalance, with the majority of flights being on-time.

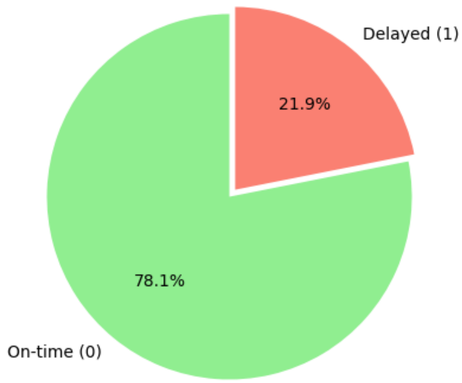

Figure 3.1: Flight Delay Distribution

3.2.2 Feature Distributions

The numerical variables show a mix of skewed and balanced distributions.

Unbalanced features:

- CRS_DEP_TIME and CRS_ARR_TIME have visible peaks during morning and evening hours, reflecting flight scheduling patterns.
- DISTANCE is right-skewed, dominated by short- to mid-range routes but with a few long-haul flights.

Balanced features:

- MONTH, DAY_OF_MONTH, and DAY_OF_WEEK are evenly distributed, showing consistent flight activity across time.
- Weather-related variables HourlyDryBulbTemperature, HourlyDewPointTemperature, HourlyRelativeHumidity, HourlyWindSpeed display near-normal patterns, indicating stable weather conditions.

The categorical features are highly imbalanced.

The variable OP_UNIQUE_CARRIER shows that Alaska Airlines (AS) operates the majority of flights, followed by SkyWest (OO) and Delta (DL).

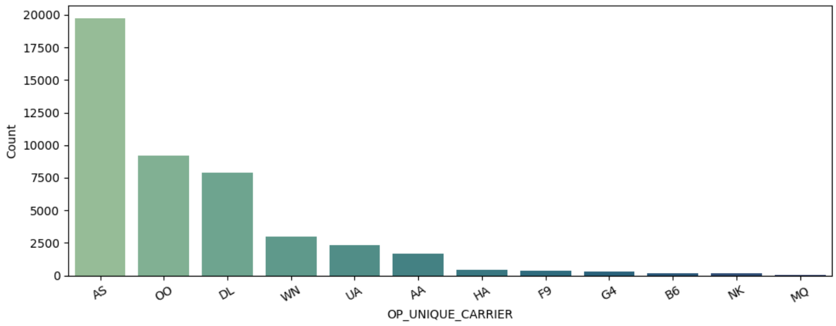

Figure 3.2: Airline Distribution

From the ORIGIN variable, most flights in Washington depart from Seattle–Tacoma Airport (SEA), accounting for about 87.41% of total flights.

The DEST_STATE_ABR distribution indicates that most destinations are in California (CA) (around 24.17%), followed by Washington (WA) and Alaska (AK).

Overall, flight activity is mainly concentrated among a few major airlines, airports, and destination states.

3.3 Bivariate Analysis with Target

3.4 Correlation and Statistical Tests

Conclusion

Through the analysis of the S&P 500 index from 1980 to 2023, this study shows how financial crises have shaped market performance, valuation, and risk. The results show that while the index demonstrates long-term growth, crisis periods are characterized by extreme volatility, sharp valuation corrections, and temporary spikes in dividend yields. These patterns confirm that financial crises create significant stress in equity markets, challenging assumptions of stability and long-term trends.

The findings provide not only a descriptive picture of historical market behavior but also practical implications. Investors can use insights from valuation ratios and volatility as early warning signals for future downturns, while policymakers may rely on these indicators to better understand systemic risks. Moreover, the derived features—such as real returns, dividend yield, and rolling volatility—create a strong base for quantitative modeling and forecasting of market resilience.

Overall, this project demonstrates the value of combining data engineering with statistical and visualization methods. By transforming historical financial data into structured insights, the study not only documents how the S&P 500 behaves under crises but also provides lessons that can support decision-making in risk management, portfolio strategy, and economic policy.

References